

Women and AI: Leveraging Language and Expression Skills in Prompt Engineering and beyond

Nisha Sethi Girdhar
AI Center of Excellence, Department of
Innovation, Fidelity International
Gurugram, India
sethinisha4@gmail.com

Abstract- This study explores the crucial role of women in prompt engineering within AI development. It emphasizes the importance of language and expression skills and how women's unique abilities offer them a fair chance in a male dominated tech area. The paper analyzes women's strengths, illustrated by experience and insights from female prompt engineers. It also provides practical advice for women aiming to venture into this field and suggests strategies to stay current with industry trends. The study concludes with a forecast on the future of women in prompt engineering and discusses measures to enhance their involvement.

Keywords—Prompt engineering, women in emerging tech, women in AI, Language skills in AI

I. PROMPT ENGINEERING AND ITS SIGNIFICANCE

A prompt is a request you give to an AI tool to get a specific response. Prompt engineering is the skill of creating and improving these requests. The goal is to ask a question or set a task clearly to get accurate and useful answers.

With increased reliance on generative AI tools in organizations, prompt engineering opportunities are booming, and jobs like this are here to stay as programming languages become more human with evolution of AI. What sets this career apart is its emphasis on language, creativity, and empathy—qualities that are frequently overlooked in conventional tech positions.

II. IMPORTANCE OF LANGUAGE AND EXPRESSION IN AI DEVELOPMENT

The evolution of AI is revolutionizing programming languages, making them more user-friendly and intuitive. This change is particularly impactful in prompt engineering. AI-enhanced programming environments can now interpret commands that resemble natural human language, reducing the complexity of coding. This not only streamlines the work of prompt engineers but also opens the field to those without extensive programming experience.

To help AI understand and respond to user queries better, it is crucial to draft clear and concise prompts. Using well-crafted language helps AI models grasp the intent behind a user's query. Understanding the nuances of language and being able to express prompts clearly and precisely is important, as it leads to more accurate and relevant responses.

Furthermore, AI tools have been known to carry and mirror societal bias, techniques like prompt engineering provide a way to balance this. Effective language helps optimize the way we communicate with these models,

ensuring that they comprehend the depth, context, and intention behind our queries [1].

III. WOMEN'S POTENTIAL IN PROMPT ENGINEERING

This is no secret that women often excel in linguistic capabilities. Several early years studies highlight that girls consistently score higher in language tests than boys. Girls tend to be more verbally fluent, and they often have larger vocabularies [2].

More recent studies examining human behavior have found that women tend to demonstrate more positive attitudes towards language learning [3]. Female students generally enjoy writing more and place greater value on language skills. One reason for this could be that boys are typically encouraged and motivated to pursue mathematics and science from an early age [3], shaping their interests in those areas. This leads to the differences we observe today.

The resultant observation is that women seemingly are well placed for prompting roles. Prompt engineering opens many opportunities in the tech field, which may otherwise feel out of reach for women.

Moreover, the unique communication skills often associated with women can be highly beneficial in refining AI prompt engineering. Their ability to utilize empathetic language and emotional intelligence can help generate more balanced and nuanced responses from AI. This approach could significantly transform the current transactional nature of AI interaction into a more collaborative dialogue. By incorporating these empathetic perspectives into prompt engineering, we can foster a more inclusive and thoughtful AI response, promoting equity in AI design and usage.

The industry now acknowledges prompting as pivotal to generative AI, exemplified by a global competition held in Dubai this year with a commitment to train a million in this field. Among the winners were Austrian Megan Fowkes in Art and underage UK prodigy, Zara, who excelled in Literature, merging her storytelling passion with AI fascination [4].

IV. THE UNIQUE PERSPECTIVES OF WOMEN IN PROMPT ENGINEERING

A view from the desks of real-life prompt engineers [6]

A. *Creativity*: The creative aspect of prompt engineering involves crafting prompts that are not only logical but also innovative. This creativity can lead to more effective and engaging prompts. Women often have strong skills in

creative problem-solving, which can be advantageous in prompt engineering.

- B. Language and Expression Skills:* Understanding language nuances, grammar, and syntax is crucial for creating clear prompts. The ability to adapt language to various contexts and audiences, and to express complex ideas effectively, is also important.
- C. Diligence:* Meticulousness is a crucial element in crafting prompts. A keen eye for detail and the ability to spot potential ambiguities or misunderstandings in prompts before they cause confusion ensuring logical consistency and clarity.
- D. Collaboration:* Prompting is a collaborative approach, different parts of the business to understand various perspectives. Women are very good at bringing everyone together, fostering an inclusive environment that encourages diverse viewpoints and innovative solutions. Their ability to facilitate communication and build strong relationships can drive successful outcomes.
- E. Patience and Persistence:* Perfecting a prompt often involves trial and error and numerous revisions. Women's nurturing skills can bring a thoughtful and considerate approach to this process, while their empathy can lead to a deeper understanding of user needs, reflected in the prompts created. Patience is required to weather this process, and persistence, often underlined by empathetic and nurturing qualities, is essential for continuous improvement.

Prompt engineering is a field that engages both analytical and creative thinking. This multidisciplinary approach allows for the development of comprehensive and effective prompts [6].

V. WHY WOMEN IN PROMPT ENGINEERING FIND THEIR CAREERS FULLFILLING

The creative process of prompt engineering is not only highly rewarding but also multifaceted and dynamic. It begins with a deep understanding of various business problems and the creation of prompts that address those issues, considering the perspectives of end-users and stakeholders to ensure that the prompts are relevant and effective.

Prompt engineers value the problem-solving aspect. The satisfaction derived from this iterative approach, seeing a prompt successfully achieve its intended outcome can be incredibly fulfilling.

Additionally, prompt engineering offers valuable collaboration and networking opportunities, allowing interaction with different parts of the business, meeting new people and providing the ability to help others. [6]

VI. ADVICE FOR ASPIRING WOMEN PROMPT ENGINEERS

- A. Build a Solid Foundation:* Though prompt engineering does not require one to be very technical, a good basic understanding of AI and large language models is a must. It is advisable to develop a solid understanding of the basics. Take advantage of online courses, try platforms like LinkedIn Learning to learn both basic and advanced techniques of prompt engineering.
- B. Develop a Problem-solving Mindset:* Start by thoroughly understanding the business problem at hand. Use analytical thinking to break it down into manageable parts and employ creativity to devise innovative solutions. Embrace an iterative approach to refining and optimizing prompts, paying meticulous attention to detail in language usage.
- C. Strong Understanding of Language:* Develop a strong command of language and understand the nuances of different words and expressions.
- D. Seek Mentors and Communities:* Find mentors or join communities to share and discuss ideas. People in these communities often are incredibly supportive of each other.
- E. Experiment on Personal Projects:* Utilize free resources such as ChatGPT, Perplexity, or Google Gemini to practice your own projects. This will help in gaining experience and becoming more comfortable with the process. For instance, you can use prompting to plan a home renovation project, allowing you to practice creating and refining complex prompts while also making progress on a personal goal.
- F. Continuous Learning:* Consistent practice is essential. Do not be discouraged if you don't achieve immediate success, as there is a natural learning curve involved. Instead, view it as a long-term commitment, like running a marathon rather than a sprint, and persist in your efforts to develop your expertise. Embrace adaptability and commit to ongoing technical development to effectively tackle new information and tools.

VII. CHALLENGES FACED BY WOMEN IN PROMPT ENGINEERING

- A. Longevity and Relevance:* As AI and other technologies progress rapidly, there might be apprehensions about whether their current abilities in prompt engineering will remain valuable and applicable in the long run. It is crucial to focus on adapting one's prompt engineering skills with time, new models and new communication approaches.
- B. Lack of Awareness of this Role:* Prompt engineer as a formal job title is not widely implemented. This lack of recognition can make it difficult for women to identify and pursue opportunities in this field, as well as hinder their ability to advocate for their work and its importance.

C. *Lack of Role Models*: Since it is a relatively nascent field, women in prompt engineering face lack of visible and accessible role models. This lack of representation can make it difficult for women to envision themselves in similar roles and hinder their ability to connect with mentors who can provide guidance and support.

VIII. TIPS FOR STAYING UPDATED IN THE FIELD

Here are some tips from real life prompt engineers:

A. *Notifications & Alerts*: There is always something happening, a new model coming out, new features etc. One of the quickest ways to stay updated is setting up alerts. Tracking updates on AI, machine learning, and prompt engineering helps to understand the latest models and how to adapt prompts for them.

For technical details on prompt engineering, there is a wealth of information available online like promptingguide.ai. For updates related to large language models (LLMs), blogs and notebooks on HuggingFace and Kaggle are recommended [5].

B. *Webinars and Conferences*: Once you find the right information sources, they are likely to offer details about webinars and conferences. Attending these events can be helpful for staying informed about the latest advancements in the field.

C. *Colleagues and Networking*: Relying on colleagues and networking within the industry to learn about new developments and share insights is beneficial. There is a constant stream of new research, tools, and techniques. Many professionals are actively experimenting in their own spaces, and discussing these experiments can spark new ideas and collaborative opportunities.

Additionally, being part of a professional network allows for access to exclusive resources, such as private forums, specialized trainings, and early access to new technologies.

IX. FUTURE OF WOMEN IN PROMPT ENGINEERING

Prompt engineering is accessible to everyone, making it a wonderful opportunity for women to get involved in emerging technology.

Women often bring skills like creativity and problem-solving to the table, which are highly beneficial in prompt engineering. Their strong empathy and high emotional intelligence make them well-suited for such roles that benefit from a deep understanding of human psychology and the dynamics of human-technology interaction. This enables them to not only excel in prompt engineering but also contribute significantly to its evolution.

There are some exceptional women in AI who have made significant contributions, such as pioneering emotional AI and creating programs like 'AI for All' to encourage girls in the field [6].

Women already use prompts in their daily lives—asking for weather updates, setting reminders, sending texts, making grocery lists, playing music, and searching online. These interactions with technology can boost their confidence to enter the field of prompt engineering. And as AI continues to

advance, it is important to recognize that it can unlock opportunities that were previously unattainable. Prompt engineering is just one of the many.

X. CONCLUSION

The value of techniques that focus on human interaction, such as prompt engineering, becomes increasingly clear with the progress of AI. By leveraging the unique skills of women, we can help to develop AI systems that are more functional, collaborative, and inclusive. Through interviews with prompt experts, we have discovered an exciting opportunity that aligns with our strengths and is particularly rewarding for women. However, the lack of awareness of prompt engineering as a formal job title and the gap in understanding its opportunities and challenges pose significant obstacles to overcome. Here are some recommendations for next steps:

- Raise awareness of prompt engineering through case studies, articles, conference presentations, and social media discussions
- Create mentorship programs and networking opportunities to connect women with experienced professionals who can offer guidance and support.
- Promote diversity and inclusion within the field to create a more supportive and inclusive environment for women in prompt engineering.
- Conduct future research to address the gap in awareness of opportunities and challenges, and identify strategies for increasing the visibility and recognition of the field.

SOURCES AND REFERENCES

- [1] Breaking the Bias: Gender Fairness in LLMs Using Prompt Engineering and In-Context Learning
<https://rupkatha.com/V15/n4/v15n410.pdf>
- [2] Sex differences in the adult brain university of edinburgh
<https://www.pure.ed.ac.uk/ws/files/59812674/bhy109.pdf>
- [3] The possible relationship between gender and motivation
FULLTEXT01.pdf
- [4] <https://www.businesswireindia.com/dubai-awards-three-winners-from-india-and-austria-at-first-of-its-kind-global-prompt-engineering-championship-90143.html>;
<https://www.sthelens.london/blog/2024/10/11/zara-from-year-9-youngest-participant-at-global-prompt-engineering-championship/>
- [5] <https://www.promptingguide.ai/> ; <https://huggingface.co/blog>;
<https://www.kaggle.com/discussions/getting-started/109256>
- [6] <https://ai-4-all.org/> ; <https://www.affectiva.com/>
- [7] Interview with Prompt Engineers - Prompt experts from various industries, including AI programming service providers, financial services, and legal sectors, were interviewed for this study. The participants consisted of a total of six individuals, five were female and one male. Among the participants, two were in the initial years of their career, while four others were experienced professionals.

Note: In the rapidly evolving tech landscape, this paper focuses specifically on women's engagement in AI through the lens of prompt engineering, highlighting it as a significant area where their unique skills can shine. However, it's essential to recognize that women have diverse opportunities beyond this niche. Fields such as data science, software development, cybersecurity, AI ethics, user experience design, and product management are increasingly welcoming women, allowing them to leverage their strengths in communication and creativity. While the paper underscores the importance of prompt engineering, it also serves as a reminder that women can excel across various sectors in the tech industry.